



Week 5 Lecture 3

▼ Class	BSCCS2003
🕒 Created	@October 4, 2021 12:51 AM
🔗 Materials	
# Module #	29
▼ Type	Lecture
☰ Week #	5

CRUD

Types of operations - Create

- Create a new entry
- Must not already exist
- Check within database to avoid conflicts
- Mention mandatory vs optional fields (name, address, mobile number) ...

Types of operations - Read

- Get a list of students
- Summarize number of students, age distribution, geographic locations
- Plot histograms of marks

Types of operations - Update

- Change of address
- Update marks
- Change start date of course

Type of operations - Delete

- Remove graduated students
- Delete mistaken entries

- Un-enroll students from course

CRUD

Create - Read - Update - Delete

- Originally in context of database operations: nothing to do with the Web
- Reflects cycle of data models
- Databases optimized for various combinations of operations
 - Read-heavy: lots of reading, very little writing or creating
 - Write-heavy: security archive logs

API - Application Programming Interface

- Standardized way to communicate with a server
- Client only needs to know the API - now how the server implements the database, for example
- CRUD is good set of functionality for a basic API
 - Usually considered the first level API to implement a web application
- Deals only with the data model life cycle - other control aspects possible